

IP address can be obtained by typing 'whatsmyip.org' in a browser.) The next step is port forwarding.

Port forwarding

Since the C program connects to the public IP (of the router), the router needs to know which local device the connection request should be forwarded to. This section demonstrates how to set up port forwarding in the NETLINK HG323DAC modem. Other routers will follow a similar procedure. In case of doubt, refer to online resources.

In a browser, type '192.168.1.1'.

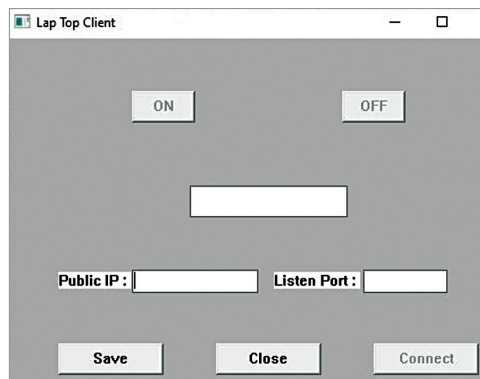


Fig. 6: UI layout

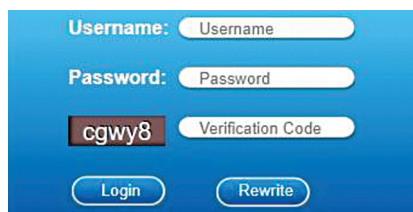


Fig. 7: Router login page

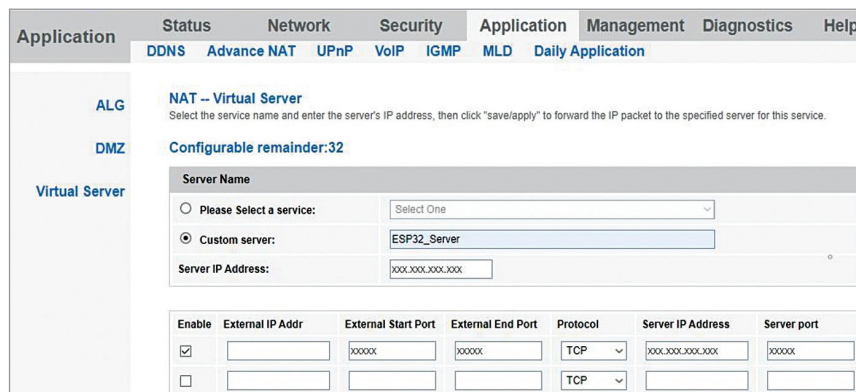


Fig. 8: Application setup

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The login page of the router will appear. Enter the credentials and click Login. Fig. 7 shows the router login page. Refer to Fig. 8 for application setup. Navigate to:

Application > Advanced NAT > Virtual Server > Add.

1. Select Custom Server and enter a name, such as ESP32_Server.
2. In the Server IP Address field, enter the local IP address of the ESP32.
3. Tick the Enable checkbox.
4. Leave the External IP Address field blank.
5. Enter the ESP32 listening port number in both the External Start Port and External End Port fields.
6. Set the Protocol to TCP.
7. Again, enter the ESP32's local IP address in the Server IP Address field.
8. Enter the ESP32 listening port number in the Server Port field.
9. Click Save/Apply.

One final step remains. Click on the DMZ option. Enable the DMZ Host feature. Enter the local IP address of the ESP32 in the DMZ Host IP Address field. Click Save/Apply to confirm.

To complete the setup, click Logout. Port forwarding is now complete, and the ESP32 can be accessed from anywhere in the world.

Operation

Begin by powering the ESP32 board, either through the USB port (e.g., using a power bank) or by supplying regulated power to the Vin pin.

Then, run the client by executing the .exe file generated from the C code.

The public IP address and the listening port number must be entered and saved. Click the Connect button. If the connection is successful, the On and Off buttons will become active, allowing remote control of the LED.

Problems

The public IP address may change each time the router is rebooted. This issue can be resolved by purchasing a static IP address from the internet service provider.

It is also essential to ensure that the IP address assigned to the ESP32 is unique within the local network to avoid IP conflicts.

This concludes the discussion on the remote control over internet system.

Testing

After completing all configurations:

1. Power on the device.
2. Run the .exe file and launch the GUI.
3. Use the On and Off buttons to control the LED.

The LED should now be controllable remotely. **EFY**



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